TrustV2X: Explainable Hybrid Learning for Trust-Aware C-ITS/V2X Safety-Message Misbehaviour Detection

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***Abstract*—Cooperative Intelligent Transportation Systems (C-ITS) and Vehicle-to-Everything (V2X) communications enable vehicles, roadside infrastructure, network entities and vulnerable road users to exchange safety-critical information for connected, autonomous and smart mobility. However, authenticated communication does not guarantee that the exchanged message content is physically or temporally trustworthy. A faulty or compromised vehicle may transmit syntactically valid but semantically false safety messages, including false position, speed, heading, acceleration or replayed mobility information. This paper proposes TrustV2X, an explainable hybrid learning framework for V2X misbehaviour detection and trust-aware safety-message validation at roadside edge nodes. TrustV2X derives temporal and mobility-consistency features from CAM/BSM-like messages, applies supervised ML/DL classifiers, explains detection decisions using feature-importance analysis, and assigns an operational trust score for accept, flag or reject decisions. The framework is evaluated on the VeReMi Extension benchmark under a vehicle-disjoint holdout, in which test vehicles are unseen during training, and is additionally validated on a controlled SUMO-based C-ITS simulation. Under the vehicle-disjoint holdout, XGBoost with TrustV2X no-rule-flags features achieved an F1-score of 0.9692 and recall of 0.9854, while the best deep-learning model, MLP, achieved an F1-score of 0.9664 with a lower false-alarm rate. Compared with raw BSM/CAM-like features, TrustV2X improved vehicle-disjoint F1-score by 43.26%. Controlled SUMO simulation, used for internal validation of the feature pipeline under known attack injection, produced F1-scores above 0.999 across random, unseen vehicle and time-based settings. These results demonstrate the value of temporal and plausibility-aware message representation for trustworthy C-ITS safety messaging.**

***Index Terms*—Cooperative Intelligent Transportation Systems (C-ITS), Vehicle-to-Everything (V2X) Communications, Misbehaviour Detection, Machine Learning, Roadside Edge Intelligence, Explainable Artificial Intelligence (XAI).**

I. Introduction

Cooperative Intelligent Transportation Systems (C-ITS) and Vehicle-to-Everything (V2X) communications are central to connected, cooperative, autonomous and intelligent mobility. In a C-ITS environment, vehicles use on-board units to broadcast safety messages to nearby vehicles, roadside infrastructure and network entities. These messages normally contain movement-related information such as position, speed, acceleration, heading and timestamp. When exchanged reliably, such information improves cooperative awareness, early hazard detection, collision avoidance, traffic efficiency and smart-city mobility services [1]–[3].

However, C-ITS safety depends on more than whether a message is transmitted successfully or cryptographically authenticated. A message may originate from a legitimate sender but still contain false or misleading content. For example, a faulty or compromised vehicle may claim an impossible location change, report abnormal speed, transmit inconsistent heading information or replay an earlier movement state. These attacks are dangerous because they target the semantic trustworthiness of safety messages rather than only the communication channel. A malicious message that appears valid at protocol level may still corrupt local situational awareness and cause unsafe downstream decisions in cooperative mobility applications [4], [5].

This makes V2X misbehaviour detection a critical research problem. Traditional mechanisms such as authentication, certificate management, and secure communication protocols are necessary, but they do not ensure that message content is correct. Data-centric detection is therefore required to assess whether reported vehicle behaviour is physically, temporally, and contextually plausible. Existing research has investigated plausibility checks, machine-learning classifiers, deep-learning models, roadside detection, and federated learning for vehicular security [6]–[11]. Nevertheless, several challenges remain. Many approaches emphasise classification accuracy, while safety-critical C-ITS deployment also requires strong recall, low false-alarm rates, robustness, low-latency inference, and interpretability. In addition, models evaluated only on internally generated data may not generalise well to external benchmark conditions or unseen vehicles.

To address these challenges, this paper proposes TrustV2X, an explainable hybrid machine-learning and deep-learning framework for detecting misbehaviour in V2X safety messages. TrustV2X is designed as a roadside-edge intelligence layer. It does not optimise vehicle routes or orchestrate traffic

flow; instead, it evaluates the trustworthiness of incoming safety messages before they influence cooperative awareness and downstream safety decisions. The framework combines temporal and mobility-consistency feature engineering, supervised learning, trust scoring, and explainability. This makes it suitable for roadside unit (RSU) deployment where detection must be fast, interpretable, and robust against false or inconsistent V2X messages.

The central argument of this paper is that raw safety-message attributes alone are not sufficient for reliable V2X misbehaviour detection. A stronger representation is needed to capture how a sender's movement evolves over time and whether the claimed behaviour is physically plausible. TrustV2X therefore derives features such as inter-message time difference, distance from previous position, estimated speed from displacement, speed–position mismatch, heading change, message rate, sender history, and spatio-temporal identity clustering. These features are evaluated using both machine-learning and deep-learning models under controlled SUMO simulation and external VeReMi benchmark validation.

This work assumes an RSU/edge-assisted C-ITS environment in which vehicles periodically broadcast CAM/BSM-like safety messages containing position, speed, acceleration, heading, and timestamp information. The attacker is assumed to control or compromise a vehicle or message source capable of transmitting syntactically valid but semantically false mobility messages. Such attacks may include false position, speed falsification, abnormal heading, replayed movement states, Sybil-like identity behaviour, or excessive message transmission. TrustV2X is not intended to replace cryptographic authentication; rather, it complements authentication by detecting physically and temporally implausible message content before it influences cooperative awareness or safety decisions.

The main contributions of this paper are fourfold. First, it proposes a practical TrustV2X framework that integrates plausibility-aware feature engineering, ML/DL-based detection, trust scoring and explainability for C-ITS safety-message validation. Second, it introduces a two-stage experimental design combining controlled SUMO-based validation with external VeReMi Extension validation. Third, it provides a balanced comparison of machine-learning and deep-learning models, showing that the strongest configurations cluster within 0.3 F1 points, with XGBoost giving the best single-run result and MLP the lowest false-alarm rate. Fourth, it discusses detection trade-offs using F1-score, recall, false alarm rate, Matthews Correlation Coefficient (MCC) and latency rather than relying on accuracy alone.

In TrustV2X, the term *hybrid* refers to the fusion of learned and rule-based evidence within the trust-scoring layer, where a model-estimated attack probability is combined with rule-violation and plausibility risk, rather than to an ensemble of machine-learning and deep-learning classifiers.

II. Related Work

Research on V2X security has shown that connected vehicles are exposed to threats such as spoofing, replay attacks, Sybil attacks, denial-of-service attacks, false data injection, and privacy leakage. Existing surveys on V2X security and C-ITS misbehaviour detection emphasise that authentication and secure communication protocols are necessary, but they cannot guarantee that transmitted message content is correct [2]–[5]. TrustV2X targets this content-trust gap by detecting whether safety messages are behaviourally plausible.

Data-centric V2X misbehaviour detection has been explored through mobility consistency, position verification, consecutive message analysis, and machine-learning classification. These approaches show that abnormal position changes, speed inconsistencies, heading deviations, unusual message timing, and local adaptive behaviour can reveal malicious or faulty vehicles [6]–[8], [12], [13]. However, many studies do not clearly separate raw message attributes from engineered temporal and plausibility-aware features. TrustV2X addresses this by comparing raw BSM/CAM-like features against TrustV2X-enhanced features, including a No-Rule-Flags variant.

Deep-learning approaches have also been proposed for VANET, IoV, and C-ITS anomaly detection. In message-level tabular datasets, a multilayer perceptron is often a better fit than convolutional or recurrent models unless the data is explicitly converted into vehicle-level temporal windows [9]–[11]. In the final TrustV2X evaluation, MLP performs strongly on VeReMi, while 1D-CNN does not generalise well under vehicle-disjoint validation.

Recent work confirms the value of engineered representations on the same benchmark. Youness *et al.* [14] report that domain-informed spatiotemporal and communication-aware features substantially outperform raw and dimensionality-reduced attributes on the VeReMi Extension dataset, while sequence models such as HyperLSTM [15] exploit vehicle-level temporal windows. These studies evaluate under random stratified splits. TrustV2X differs by adopting a vehicle-disjoint protocol, by separating rule-derived from continuous features through the No-Rule-Flags variant, and by coupling detection with an operational trust decision rather than a binary label.

Edge-based and federated vehicular security models are relevant because V2X detection must operate close to where messages are received. Roadside units and edge nodes can observe local message streams and support low-latency decision-making; federated learning can further support privacy-preserving distributed training [16], [17]. TrustV2X follows this roadside-edge direction but keeps the current design lightweight and deployable.

Explainability is also important in safety-critical mobility systems: operators and downstream applications need to understand why a message was flagged. TrustV2X therefore includes an explainability layer that highlights the most influential temporal and mobility-consistency features [18], [19].

Simulation and benchmark datasets remain essential because malicious real-world V2X datasets are difficult to collect safely and at scale. SUMO supports controlled scenario generation [20], while the VeReMi dataset family supports benchmark evaluation of V2X misbehaviour detection [21], [22]. TrustV2X uses both to combine internal validity with external benchmark generalisation.

III. TrustV2X Framework

TrustV2X is designed as a roadside edge-based misbehaviour-detection and trust-management framework for C-ITS safety messaging. Fig. 1 shows TrustV2X as an RSU/edge security intelligence layer that evaluates incoming CAM/BSM-like messages before they are accepted, flagged or rejected for downstream C-ITS safety applications. This positioning allows the framework to complement cryptographic authentication by checking whether authenticated message content is physically and temporally plausible before it influences cooperative awareness, hazard warnings or traffic-safety decisions.

Fig. 2 summarises the proposed TrustV2X processing pipeline. Incoming CAM/BSM-like safety messages are first preprocessed to extract basic message attributes, including sender identifier, timestamp, position, speed, acceleration, heading and message type. Since several V2X attacks cannot be detected reliably from a single message alone, TrustV2X also compares consecutive messages from the same sender to derive temporal and mobility-consistency features. These features capture whether the reported movement is physically plausible over time before the message is passed to the ML/DL detection, trust-scoring and explainability layers.

For a message i received at time t_i, where the previous message from the same sender was received at t_{i-1}, the inter-message time difference is defined as

$$\Delta t_i = t_i - t_{i-1}. \tag{1}$$

The spatial displacement between two consecutive reported positions is computed as

$$d_i = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}, \tag{2}$$

where (x_i, y_i) represents the reported position of the current message and (x_{i-1}, y_{i-1}) represents the previous reported position. From this displacement, TrustV2X estimates the implied movement speed as

$$\hat{v}_i = \frac{d_i}{\Delta t_i}. \tag{3}$$

The difference between the reported speed and the speed estimated from positional displacement is calculated as

$$E_i^{speed} = |\hat{v}_i - v_i|, \tag{4}$$

where v_i is the speed reported in the current safety message. A high value of E_i^{speed} may indicate false position reporting, speed falsification, replayed movement, or inconsistent mobility behaviour.

Heading consistency is also important because sudden heading changes may be physically unrealistic depending on speed and time interval. The heading difference is computed using circular angular distance:

$$\Delta\theta_i = \min\big(|\theta_i - \theta_{i-1}|, 360^\circ - |\theta_i - \theta_{i-1}|\big), \tag{5}$$

where θ_i and θ_{i-1} are the current and previous headings (in degrees). TrustV2X also considers message-rate behaviour. For sender s, the number of messages transmitted within a time window W is represented as

$$R_s(W) = \frac{N_s(W)}{|W|}, \tag{6}$$

where $N_s(W)$ is the number of messages from sender s observed within window W, and $|W|$ is the window duration.

The framework separates features into three groups. Raw BSM/CAM-like features contain original message attributes such as timestamp, position, speed, acceleration, and heading. TrustV2X No-Rule-Flags features add continuous temporal and mobility-consistency features but exclude binary rule flags. TrustV2X Full features include both continuous TrustV2X features and binary plausibility indicators. This separation tests whether performance comes from a general behavioural representation rather than simply reproducing hand-coded rules.

The detection layer applies supervised machine-learning and deep-learning models to classify each message as normal or malicious. The machine-learning models include Logistic Regression, Random Forest, XGBoost [23], and LightGBM [24]. The deep-learning models include MLP and 1D-CNN. LSTM is excluded because this experiment uses message-level tabular records rather than true sequential windows. A valid LSTM model would require each sample to be represented as a sequence of consecutive messages per vehicle.

The trust-scoring layer converts detection evidence into an operational trust value. Let $P_{\text{attack}}(i)$ be the model-estimated probability that message i is malicious. Let R_i^{rule} represent rule-violation risk and R_i^{plaus} represent plausibility risk. The combined risk score is defined as

$$R_i = \alpha P_{\text{attack}}(i) + \beta R_i^{\text{rule}} + \gamma R_i^{\text{plaus}}, \tag{7}$$

where α, β, and γ are tunable weights satisfying $\alpha+\beta+\gamma = 1$ and $\alpha, \beta, \gamma \geq 0$. The final trust score is then computed as

$$T_i = 1 - R_i. \tag{8}$$

In the controlled trust-scoring experiment, the weights were set to $\alpha = 0.65$, $\beta = 0.20$ and $\gamma = 0.15$. This gives the highest contribution to the classifier output while still retaining rule-violation and plausibility evidence. The plausibility risk R_i^{plaus} is derived from continuous temporal and mobility-consistency indicators, including speed–position mismatch, distance jump, heading change, speed change, message-rate behaviour, identity clustering and time-gap irregularity. The rule-violation risk R_i^{rule} is derived from binary plausibility indicators such as unrealistic speed, hard acceleration, large

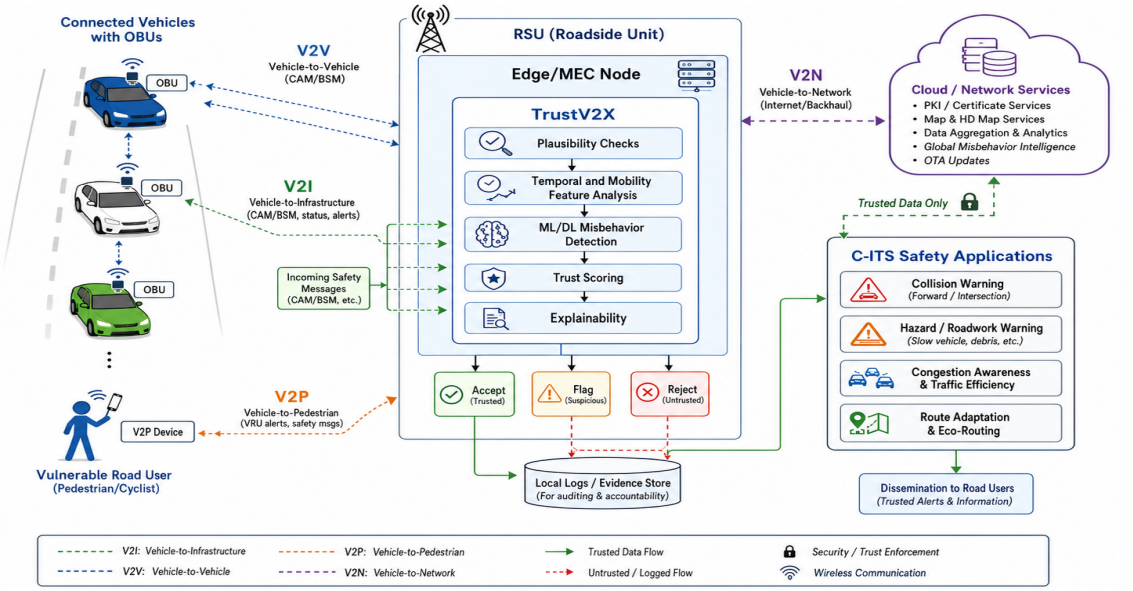


Figure 1. TrustV2X positioning within a C-ITS/V2X roadside-edge architecture.

position jump, speed–position mismatch, heading anomaly, irregular timestamp, abnormal message rate and possible Sybil-like behaviour.

The decision policy is defined using accept and reject thresholds:

$$\text{Decision}_i = \begin{cases} \text{Accept}, & T_i \geq \tau_{\text{accept}}, \\ \text{Flag}, & \tau_{\text{reject}} \leq T_i < \tau_{\text{accept}}, \\ \text{Reject}, & T_i < \tau_{\text{reject}}. \end{cases} \tag{9}$$

For the reported implementation, the thresholds were set to $\tau_{\text{accept}} = 0.75$ and $\tau_{\text{reject}} = 0.40$. Therefore, TrustV2X provides more than a binary classifier: it gives the C-ITS application an interpretable decision that can accept trusted messages, flag uncertain messages, or reject suspicious messages before they influence safety-critical operations.

IV. Experimental Design

The experimental evaluation assesses whether TrustV2X improves V2X misbehaviour detection beyond raw message attributes and whether performance generalises beyond controlled simulation. The study uses two complementary datasets: a controlled SUMO-based C-ITS simulation and the VeReMi Extension benchmark.

The controlled SUMO experiment uses the 4x4 traffic scenario shown in Fig. 3 to generate CAM/BSM-like safety-message traces for internal TrustV2X validation. The resulting dataset contains 40,000 safety-message records, comprising 28,000 normal messages and 12,000 attack messages. The injected attacks include false position, gradual position drift, speed falsification, low-speed spoofing, heading falsification, replay attack, sudden jump, ghost vehicle, Sybil attack, intermittent attack, DoS flooding and neighbour inconsistency. This controlled setting provides a repeatable environment in which the attack types and message-generation process are known before the framework is further evaluated using the external VeReMi benchmark.

Table I
Dataset profiles for controlled SUMO and VeReMi Extension

Property	Controlled SUMO	VeReMi Extension
Total records	40,000	250,000
Normal records	28,000	167,363
Attack records	12,000	82,637
Attack categories	12	5
Validation role	Internal	External

The VeReMi Extension benchmark provides a stronger external validation setting. The processed VeReMi dataset contains 250,000 labelled records, consisting of 167,363 normal messages and 82,637 attack messages. The attack categories include constant position, constant speed, replay attack, DoS attack, and DoS-Sybil attack. For the main VeReMi evaluation, binary labels were derived from the dataset attack-scenario metadata: messages with *attack_type* set to normal were labelled as benign, while messages from attack scenarios were labelled as malicious. This was verified by comparing the final label column against the *attack_type* field, with no mismatches observed. Ground-truth mobility fields, derived physical error values, message identifiers, sender identifiers, and source-file indicators were excluded from all model inputs to prevent data leakage.

The VeReMi evaluation uses random stratified holdout and vehicle-disjoint holdout. The random stratified holdout preserves class balance while randomly splitting the dataset into training and testing sets. The vehicle-disjoint holdout is more challenging because the vehicles in the test set are not seen during training. This setting matters most because it tests whether TrustV2X generalises to unseen senders rather than memorising vehicle-specific patterns.

Table I summarises the two datasets used in the evaluation.

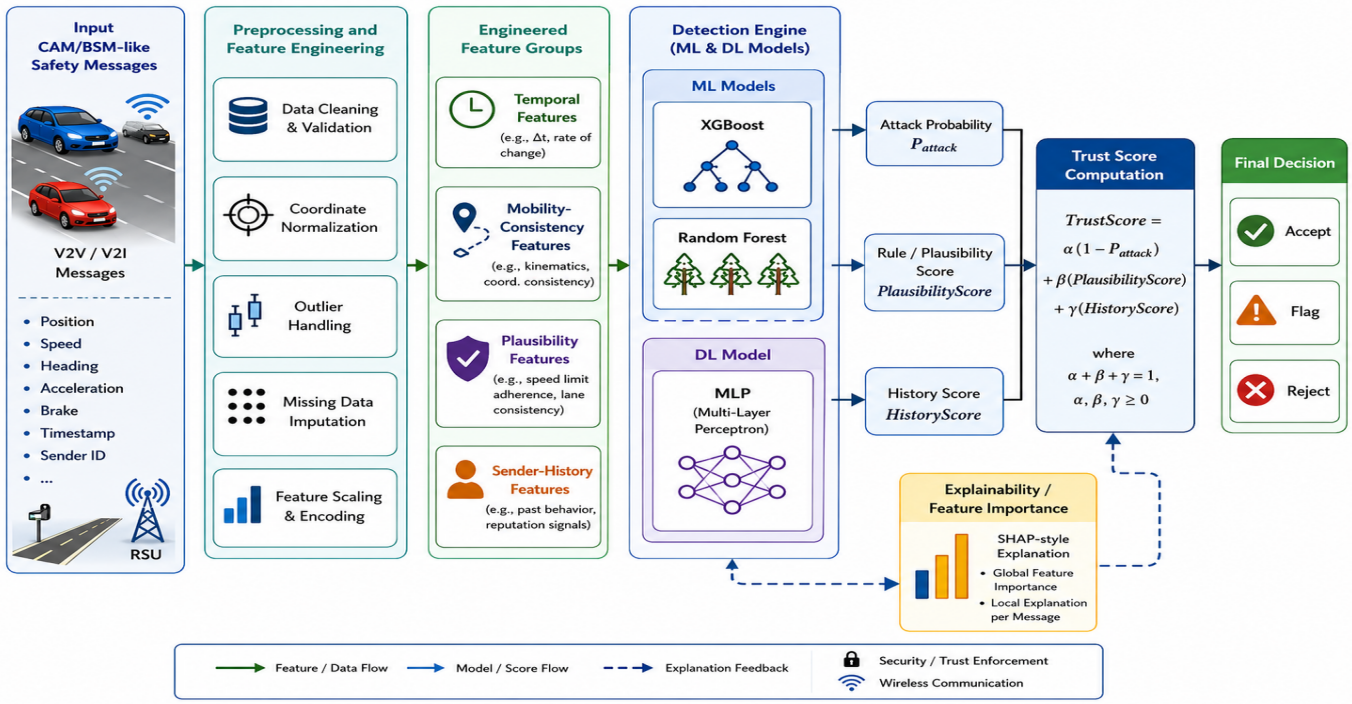


Figure 2. TrustV2X framework pipeline for V2X safety-message misbehaviour detection.

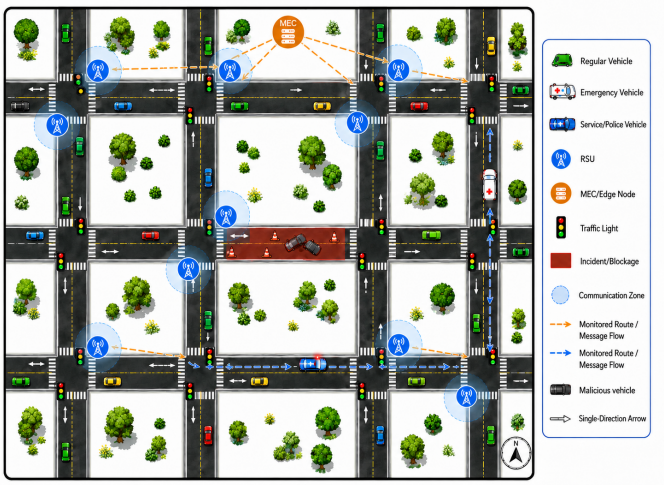


Figure 3. Controlled SUMO 4x4 C-ITS simulation scenario.

The controlled SUMO dataset provides an internal validation setting with known attack injection, while the VeReMi Extension benchmark provides external validation under established V2X misbehaviour scenarios.

The implementation was developed in Python using scikit-learn for classical machine-learning models, XGBoost and LightGBM for gradient-boosted models, and TensorFlow/Keras for deep-learning models [23]–[25]. The evaluated ML models were Logistic Regression, Random Forest, XGBoost, and LightGBM, while the DL models were MLP and 1D-CNN. For each message, TrustV2X derived the temporal and mobility-consistency features defined in Section III.

For reproducibility, XGBoost used `n_estimators`=220, `max_depth`=5, learning rate 0.05, and 0.85 subsampling/column sampling. The MLP used three ReLU hidden layers of 128, 64 and 32 units with batch normalisation, dropout and Adam optimisation. Both random and vehicle-disjoint evaluations used a 75/25 train–test split; the VeReMi vehicle-disjoint split used 159 training vehicles and 53 testing vehicles with no vehicle overlap.

Latency was measured as amortised inference time by dividing the total batch prediction time by the number of test messages. Therefore, the reported latency values represent average per-message prediction cost under batch evaluation, rather than full end-to-end real-time communication delay. This excludes wireless transmission delay, packet loss, RSU scheduling overhead and application-layer response time. All experiments were conducted on a workstation running Windows 11 and Python 3.13.7. Model training and inference were performed using CPU-based execution unless otherwise stated.

A. Evaluation Metrics

The evaluation uses binary classification, where malicious or misbehaving messages are treated as the positive class and normal messages as the negative class. True positives (TP) represent malicious V2X messages correctly detected as malicious, true negatives (TN) represent normal messages correctly classified as normal, false positives (FP) represent normal messages incorrectly flagged as malicious, and false negatives (FN) represent malicious messages incorrectly accepted as normal. The reported metrics are accuracy, preci-

Table II
CONTROLLED SUMO VALIDATION RESULTS

Setting	Acc.	Recall	F1	FAR	MCC
Random holdout	0.9999	0.9997	0.9998	0.0000	0.9998
Unseen vehicle	0.9996	0.9994	0.9994	0.0003	0.9991
Time-based	0.9998	0.9993	0.9997	0.0000	0.9995
Hard ablation	0.9852	0.9687	0.9752	0.0077	0.9647
Trust scoring	0.9997	1.0000	0.9995	0.0004	0.9993

sion, recall/detection rate, F1-score, false alarm rate (FAR), Matthews Correlation Coefficient (MCC), relative F1-score improvement and amortised inference latency.

Although accuracy is included for completeness, the evaluation prioritises F1-score, recall, FAR, MCC and latency because V2X misbehaviour detection is safety-critical. Missed malicious messages may affect cooperative awareness and safety decisions, while excessive false alarms may reduce operational usability at the RSU/edge layer. F1-score captures the balance between precision and recall, FAR measures how often normal messages are incorrectly flagged, and MCC provides a robust summary of binary classification quality. Relative improvement is reported against the best raw BSM/CAM-like baseline, while latency is reported as amortised per-message prediction time under batch evaluation.

V. RESULTS AND DISCUSSION

The experimental results show that TrustV2X provides strong misbehaviour-detection performance across both controlled SUMO simulation and external VeReMi validation. The strongest evidence is provided by the vehicle-disjoint VeReMi setting, which tests whether the framework generalises to unseen vehicles rather than memorising sender-specific patterns.

In the controlled SUMO experiment, TrustV2X achieved near-perfect performance across random, unseen-vehicle, and time-based validation settings, as shown in Table II. The random stratified holdout achieved an accuracy of 0.9999, recall of 0.9997, F1-score of 0.9998, false alarm rate of 0.0000, and MCC of 0.9998. The unseen-vehicle split achieved an F1-score of 0.9994, while the time-based split achieved an F1-score of 0.9997. These results show that the framework detects a wide range of simulated V2X misbehaviour patterns when the environment and attack-injection process are controlled.

The hard ablation test provides a more conservative interpretation of the controlled SUMO results. When timing-heavy and binary rule-flag features were removed, TrustV2X still achieved an F1-score of 0.9752, a false alarm rate of 0.0077, and an MCC of 0.9647. This suggests that the framework is not solely dependent on simple rule flags or timing artefacts. Instead, it captures broader movement and message-consistency patterns that remain useful even when some strong engineered features are removed. The operational trust-scoring experiment also supports the practical deployment argument, achieving recall of 1.0000, F1-score of 0.9995 and false alarm rate of 0.0004 in the controlled setting.

The VeReMi external benchmark provides the stronger test of generalisation. Under the vehicle-disjoint holdout, the best

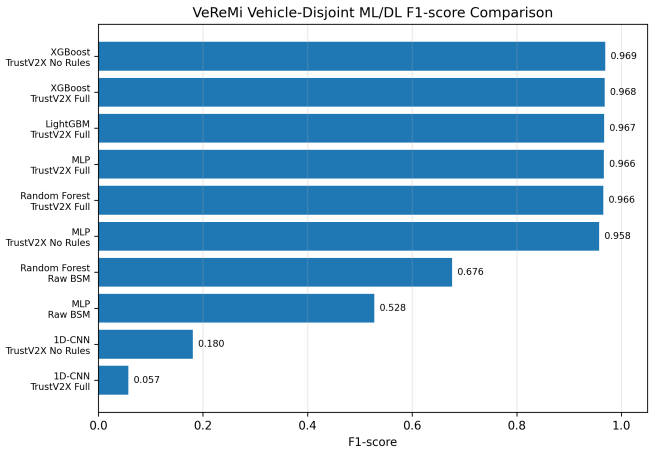


Figure 4. VeReMi vehicle-disjoint F1-score comparison across ML/DL models.

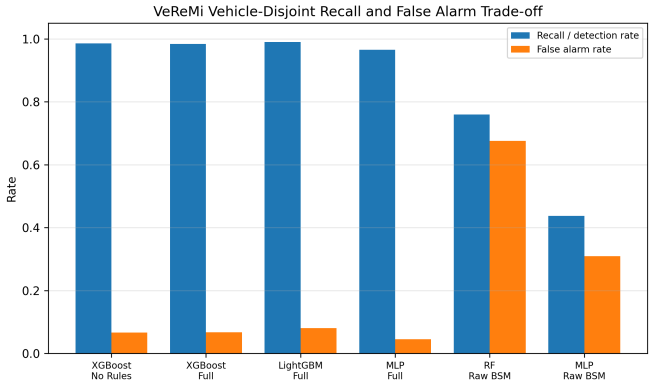


Figure 5. VeReMi recall and false-alarm rate trade-off.

overall result was achieved by XGBoost using TrustV2X No-Rule-Flags features. This model achieved an accuracy of 0.9636, precision of 0.9536, recall of 0.9854, F1-score of 0.9692, false alarm rate of 0.0667, MCC of 0.9254, and average latency of 0.0072 ms per message, as shown in Table III. This is the main external benchmark result of the paper.

The best deep-learning result was achieved by MLP using the TrustV2X Full feature set. It achieved an accuracy of 0.9608, precision of 0.9678, recall of 0.9650, F1-score of 0.9664, false alarm rate of 0.0451, MCC of 0.9193 and average latency of 0.0095 ms per message. This result is important because it shows that TrustV2X also supports strong neural detection, even though the best overall model is tree-based. The MLP result is very close to XGBoost and has a lower false alarm rate, making it an important model for the final discussion.

By contrast, raw BSM/CAM-like features generalised poorly under vehicle-disjoint validation. The best raw baseline, Random Forest with raw features, achieved only 0.6765 F1-score and a false-alarm rate of 0.6754. TrustV2X improved the vehicle-disjoint F1-score from 0.6765 to 0.9692, representing a relative improvement of 43.26%, while redu-

Table III
SELECTED VEREMI ML/DL VEHICLE-DISJOINT RESULTS

Family	Feature set	Model	Acc.	Prec.	Recall	F1	FAR	MCC	Amort. lat. ms
ML	TrustV2X No Rule Flags	XGBoost	0.9636	0.9536	0.9854	**0.9692**	0.0667	**0.9254**	0.0072
ML	TrustV2X Full	XGBoost	0.9625	0.9530	0.9841	0.9683	0.0675	0.9232	0.0053
ML	TrustV2X Full	LightGBM	0.9606	0.9448	0.9901	0.9669	0.0805	0.9197	0.0120
DL	TrustV2X Full	MLP	**0.9608**	**0.9678**	0.9650	**0.9664**	**0.0451**	0.9193	0.0095
ML	TrustV2X Full	Random Forest	0.9589	0.9430	0.9890	0.9655	0.0831	0.9163	0.0187
DL	TrustV2X No Rule Flags	MLP	0.9494	0.9395	0.9763	0.9575	0.0884	0.8960	0.0103
ML	Raw BSM Features	Random Forest	0.5776	0.6099	0.7595	0.6765	0.6754	0.0928	0.0532
DL	Raw BSM Features	MLP	0.5428	0.6658	0.4373	0.5278	0.3087	0.1303	0.0153
DL	Raw BSM Features	1D-CNN	0.4876	0.9348	0.1324	0.2320	0.0130	0.2136	0.0141
DL	TrustV2X No Rule Flags	1D-CNN	0.4729	0.9874	0.0993	0.1805	0.0018	0.2043	0.0165
DL	TrustV2X Full	1D-CNN	0.4323	0.9698	0.0296	0.0574	0.0013	0.1054	0.0163

[Bold denotes the best overall ML model (XGBoost, TrustV2X No-Rule-Flags) and the best DL model (MLP, TrustV2X Full) selected for discussion]

Table IV
VEREMI PER-ATTACK DETECTION PERFORMANCE

Split	Attack type	Detection rate
Random	Constant speed	0.9989
Random	Constant position	0.9967
Random	DoS-Sybil	0.9856
Random	Replay	0.9855
Random	DoS	0.9842
Vehicle-disjoint	Constant speed	0.9983
Vehicle-disjoint	Constant position	0.9894
Vehicle-disjoint	Replay	0.9848
Vehicle-disjoint	DoS	0.9829
Vehicle-disjoint	DoS-Sybil	0.9781

cing the false-alarm rate by 0.6087. Fig. 4 illustrates this trend across the selected ML and DL models.

Fig. 5 shows the trade-off between attack detection rate and false alarm rate under VeReMi vehicle-disjoint validation. This comparison is important because a practical C-ITS detector must identify malicious messages while avoiding excessive rejection of legitimate safety messages. The MLP result is notable because it provides a lower false alarm rate than the best XGBoost model, although XGBoost remains the strongest overall model by F1-score and MCC.

The 1D-CNN results were weaker because the dataset is message-level and tabular rather than a true temporal sequence. Although 1D-CNN produced high precision in some settings, it missed many malicious messages under vehicle-disjoint validation; for example, with TrustV2X Full features it achieved only 0.0296 recall and 0.0574 F1-score. Therefore, 1D-CNN is reported for completeness but not selected.

Per-attack evaluation further confirms the strength of TrustV2X. Under vehicle-disjoint validation, constant-speed attacks achieved a detection rate of 0.9983, constant-position attacks achieved 0.9894, replay attacks achieved 0.9848, DoS attacks achieved 0.9829 and DoS-Sybil attacks achieved 0.9781, as shown in Table IV. This means all VeReMi attack categories were detected above 97.8% in the most challenging external setting. Constant-speed and constant-position attacks were easiest to detect because they create strong physical inconsistencies, while DoS-Sybil and replay attacks were slightly more challenging but still detected reliably.

The explainability analysis in Fig. 6 supports the design logic of TrustV2X. The most influential features were temporal and mobility-consistency indicators, including inter-message

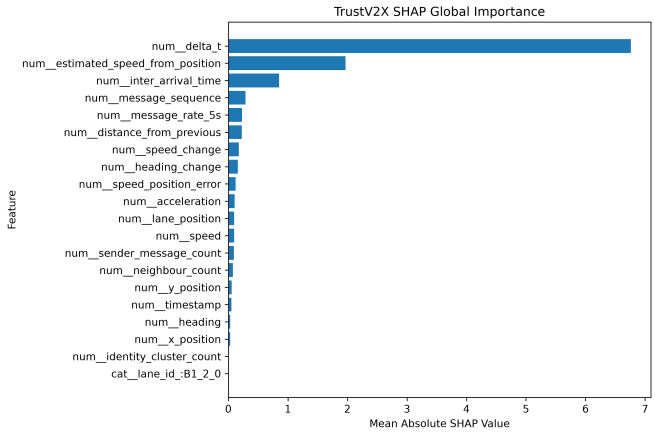


Figure 6. SHAP-based explainability analysis showing the most influential TrustV2X features for XGBoost-based V2X misbehaviour detection.

time difference, estimated speed from positional displacement, inter-arrival time, message sequence, distance from previous position, speed change, heading change, speed–position error, and acceleration. This suggests that the model is not simply memorising raw position or speed values; rather, it is learning whether the reported vehicle movement is physically and temporally plausible. Overall, the key takeaway is that strong V2X misbehaviour detection depends more on meaningful temporal and plausibility-aware message representation than on model complexity alone.

VI. LIMITATIONS

These results are based on simulation and benchmark traces, namely controlled SUMO and the VeReMi Extension benchmark, so they provide experimental evidence rather than direct on-road deployment performance. Real C-ITS/V2X deployments introduce additional uncertainty, including packet loss, channel congestion, sensor noise, pseudonym changes, heterogeneous driving styles and adaptive adversaries.

For the VeReMi evaluation, binary labels were derived from dataset attack-scenario metadata, where normal scenarios were labelled as benign and attack scenarios were labelled as malicious. Although leakage-prone fields were excluded from all model inputs, the benchmark design and attack-generation assumptions may still influence the observed performance.

Under vehicle-disjoint testing, the best model still produced a false-alarm rate of 0.0667, suggesting that deployment may require adaptive thresholds or cross-RSU confirmation. Although the SHAP-based explainability analysis provides useful feature-level insight, future work should further validate TrustV2X using vehicle-level temporal windows, sequence models and realistic roadside or network-simulation testbeds.

TrustV2X was not evaluated against adaptive adversaries. Adversarially robust V2X misbehaviour detection has been explored using generative approaches [26]; extending TrustV2X in this direction remains future work.

VII. Conclusion and Future Work

This paper proposed TrustV2X, an explainable hybrid machine-learning and deep-learning framework for trust-aware misbehaviour detection in C-ITS/V2X safety messaging. TrustV2X is positioned as an RSU/edge trust-management module that validates incoming CAM/BSM-like messages before they influence cooperative awareness and downstream safety decisions.

The framework was evaluated using controlled SUMO simulation and the VeReMi Extension benchmark. In controlled SUMO validation, TrustV2X achieved F1-scores above 0.999 across random, unseen-vehicle and time-based settings. In external VeReMi vehicle-disjoint validation, XGBoost with TrustV2X No-Rule-Flags features achieved a 0.9692 F1-score and 0.9854 recall, while MLP with TrustV2X Full features achieved a 0.9664 F1-score with a lower false-alarm rate.

Overall, the results show that temporal and plausibility-aware message representation is the main driver of performance gains over raw BSM/CAM-like attributes. The explainability analysis further supports this finding by showing that the most influential features relate to timing, movement consistency, message sequence behaviour and speed–position mismatch. Future work will extend TrustV2X with sequence-based modelling, adaptive trust thresholds, multi-RSU validation and deployment-oriented testing.

References

[1] ETSI, "Intelligent transport systems (its); vehicular communications; basic set of applications; part 2: Specification of cooperative awareness basic service," European Telecommunications Standards Institute, Tech. Rep. ETSI EN 302 637-2 V1.4.1, 2019.

[2] T. Yoshizawa, D. Singelée, J. T. Mühlberg, S. Delbruel, A. Taherkordi, D. Hughes, and B. Preneel, "A survey of security and privacy issues in v2x communication systems," *ACM Computing Surveys*, 2023.

[3] R. Sedar, C. Kalalas, F. Vázquez-Gallego, L. Alonso, and J. Alonso-Zarate, "A comprehensive survey of v2x cybersecurity mechanisms and future research paths," *IEEE Open Journal of the Communications Society*, vol. 4, pp. 325–391, 2023.

[4] A. Boualouache and T. Engel, "A survey on machine learning-based misbehavior detection systems for 5g and beyond vehicular networks," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1128–1172, 2023.

[5] M. A. Amanullah, S. W. Loke, M. B. Chhetri, and R. Doss, "A taxonomy and analysis of misbehaviour detection in cooperative intelligent transport systems: A systematic review," *ACM Computing Surveys*, vol. 56, no. 1, pp. 1–38, 2024.

[6] P. Sharma and H. Liu, "A machine-learning-based data-centric misbehavior detection model for internet of vehicles," *IEEE Internet of Things Journal*, vol. 8, no. 6, pp. 4991–4999, 2021.

[7] S. Ercan, M. Ayaida, and N. Messai, "Misbehavior detection for position falsification attacks in vanets using machine learning," *IEEE Access*, vol. 10, pp. 1893–1904, 2022.

[8] A. Sharma and A. Jaekel, "Machine learning based misbehaviour detection in vanet using consecutive bsm approach," *IEEE Open Journal of Vehicular Technology*, vol. 3, pp. 1–14, 2022.

[9] T. Alladi, B. Gera, A. Agrawal, V. Chamola, and F. R. Yu, "Deepadv: A deep neural network framework for anomaly detection in vanets," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 11, pp. 12 013–12 023, 2021.

[10] T. Alladi, V. Kohli, V. Chamola, and F. R. Yu, "A deep learning based misbehavior classification scheme for intrusion detection in cooperative intelligent transportation systems," *Digital Communications and Networks*, 2023.

[11] S. Park, D. Kim, and S. Lee, "Enhancing v2x security through combined rule-based and dl-based local misbehavior detection in roadside units," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 5, pp. 656–668, 2024.

[12] R. Sultana, J. Grover, M. Tripathi, and P. Sharma, "La-detects: Local and adaptive data-centric misbehavior detection framework for vehicular technology security," *IEEE Open Journal of Vehicular Technology*, 2024.

[13] M. F. Yuce, M. A. Ertürk, and M. A. Aydin, "Misbehavior detection with spatio-temporal graph neural networks," *Computers & Electrical Engineering*, vol. 116, p. 109198, 2024.

[14] N. Youness, A. Mostafa, M. A. Sobh, A. M. Bahaa, and K. Nagaty, "VeMisNet: Enhanced feature engineering for deep learning-based misbehavior detection in vehicular ad hoc networks," *Journal of Sensor and Actuator Networks*, vol. 14, no. 5, pp. 1–36, 2025.

[15] T. R. Mahesh, G. Gangadevi, K. K. Mishra, A. Algarni, O. Saidani, and A. Kumar, "HyperLSTM for anomaly detection model for VANET security using the VeReMi extension dataset," *International Journal of Distributed Sensor Networks*, vol. 2025, no. 6789771, pp. 1–15, 2025.

[16] A. Selamnia, B. Brik, S.-M. Senouci, A. Boualouache, and S. Hossain, "Edge computing-enabled intrusion detection for c-v2x networks using federated learning," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2022.

[17] C.-K. Tham, L. Yang, A. Khanna, and B. Gera, "Federated learning for anomaly detection in vehicular networks," in *Proc. IEEE Vehicular Technology Conference (VTC-Spring)*, 2023.

[18] S. Nazat, O. Arreche, and M. Abdallah, "On evaluating black-box explainable ai methods for enhancing anomaly detection in autonomous driving systems," *Sensors*, vol. 24, no. 11, p. 3515, 2024.

[19] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.

[20] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. L"ucken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using sumo," in *Proc. IEEE Intelligent Transportation Systems Conference (ITSC)*, 2018, pp. 2575–2582.

[21] R. W. van der Heijden, T. Lukaseder, and F. Kargl, "Veremi: A dataset for comparable evaluation of misbehavior detection in vanets," in *Proc. International Conference on Security and Privacy in Communication Systems (SecureComm)*, 2018, pp. 318–337.

[22] J. Kamel, M. R. Ansari, J. Petit, A. Kaiser, I. Ben Jemaa, and P. Urien, "Veremi extension: A dataset for comparable evaluation of misbehavior detection in vanets," in *Proc. IEEE International Conference on Communications Workshops (ICC Workshops)*, 2020.

[23] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[24] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "Lightgbm: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 3146–3154.

[25] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[26] M. H. Shahriar, M. R. Ansari, J.-P. Monteuuis *et al.*, "VehiGAN: Generative adversarial networks for adversarially robust V2X misbehavior detection systems," *ACM Transactions on Cyber-Physical Systems*, vol. 9, no. 3, pp. 1–26, 2025.